# Jeevan Thapa

**Phone:** 585-290-2847 **Email:** jt4812@g.rit.edu **Location:** Rochester, NY
**LinkedIn:** thapa-jeevan **Website:** thapajeevan.com.np **Scholar:** Google Scholar

Ph.D. candidate specializing in continual learning, parameter-efficient adaptation, and user modeling. First author of papers at ICML and ECIR, with industry research experience developing continual user-modeling methods for recommender systems at Zillow.

## RESEARCH FOCUS

Continual Learning and Generalized Category Discovery | Parameter-Efficient Adaptation | Recommender Systems and User Modeling | Gene-Disease Association

## EDUCATION

**Ph.D. in Computing and Information Sciences** *Aug 2022–Feb 2028 (Expected)*
Rochester Institute of Technology Rochester, NY
GPA: 3.97/4.00
*Relevant Coursework:* Deep Learning, Statistical Machine Learning, Non-Convex Optimization for Modern Machine Learning

**Bachelor's Degree in Computer Engineering** *Nov 2015–Sep 2019*
Pulchowk Campus, Tribhuvan University Lalitpur, Nepal
*Relevant Coursework:* Data Mining, Artificial Intelligence, Big Data Analytics, Probability and Statistics

## PUBLICATIONS

**J. Thapa**, R. Li. *Bayesian Adaptation of Network Depth and Width for Continual Learning*. In Proceedings of the 41st International Conference on Machine Learning (**ICML**), 2024.

**J. Thapa**, S. Zhao, K. Shindo. *Evolving Mixture of Low-Rank Experts for Continual User Modeling*. In Proceedings of the European Conference on Information Retrieval (**ECIR**), 2026.

**J. Thapa**, R. Li. Cross-Task Representation Alignment for Exemplar-Free Class-Incremental Learning. *Under review*.

## RESEARCH AND INDUSTRY EXPERIENCE

**Graduate Research Assistant** *Aug 2022 – Present*
Rochester Institute of Technology Rochester, NY

- Built a Bayesian continual learning framework that dynamically adapts network depth and width for evolving tasks; published at **ICML 2024**.
- Developed a cross-task representation-alignment framework that improved average accuracy by 2.16 percentage points for exemplar-free class-incremental learning; manuscript under review.
- Designed a parameter-efficient adaptation method for continual generalized category discovery using full-covariance Gaussian prototypes and evaluating the approach on medical imaging datasets.
- Developing a Bayesian adaptive graph neural network for gene-disease association prediction over protein-protein interaction graphs; currently conducting benchmark evaluation.

**Research Scientist Intern** *May 2025 – Sep 2025*
Zillow Group Remote

- Developed a mixture-of-rank-1-experts architecture for continual user modeling, enabling parameter-efficient adaptation to sequential recommendation data; published at **ECIR 2026**.

**Machine Learning Engineer** *Sep 2019 – Jun 2022*
Fusemachines Kathmandu, Nepal

- Led the development of a multimodal machine-learning pipeline for identifying potential trafficking activity in online advertisements using video, image, and text data.
- Built image–text contrastive models for advertisement matching, face-based identity linking, and BERT-based social-handle extraction, improving cross-ad linkage accuracy by 35%.
- Developed a lightweight object detector that increased inference throughput by 47%, then deployed it on NVIDIA Jetson Nano devices for real-time waste-type and disposal-intent classification.
- Authored computer-vision and time-series instructional materials for the Fusemachines AI Education Program, supporting the training of 1,000+ junior engineers.

## SKILLS

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| **Programming** | Python, C++, C |
| **ML Frameworks** | PyTorch, TensorFlow, scikit-learn, Transformers, Transformers4Rec |
| **Data and Experimentation** | NumPy, Pandas, NVTabular, MLflow, Matplotlib, Seaborn |
| **Tools and Platforms** | AWS, Git, Linux, Bash, Jira, LaTeX |

## AWARDS AND HONORS

- **RIT Ph.D. Scholarship** — Full financial support through NSF- and NIH-funded research projects.
- **Tribhuvan University Full Scholarship** — Granted by the Government of Nepal after ranking 11$^{th}$ nationwide in the entrance examination (4% acceptance rate).
- **Fusemachines AI Fellowship** — Awarded a merit-based scholarship through a nationwide selection process to pursue the AI MicroMasters program.

## TALKS

- **Poster Presentation** — "Bayesian Adaptation of Network Depth and Width for Continual Learning" at **ICML 2024**, Vienna, Austria.
- **Guest Lecture** — "Continual Test-time Adaptation" and "Deep Learning Toolkit" in graduate course CISC-865: Deep Learning, Rochester Institute of Technology.

## ACADEMIC SERVICES

- Conference reviewer for ICLR 2025–2026, ECCV 2026, and NeurIPS 2026.